

10 Billion Records vs. The Regulator

An Architectural Post-Mortem of a National Bank Crash

Martin Ladecký · Makelt & JCon Slovenia 2026



WHY ME

I WAS THERE

Experience

Tech Lead | Commerzbank

Organizer | Czech JUG

ex-CTO | Loono

Why I am telling this story

15+ years in high-stakes banking and NPO digital transformation. Currently architecting the "Exit Strategy" for massive European data volumes.

Known for "war stories" from critical system failures and regulatory audits.

A candid, practical, and highly creative problem solver.

Part I: The Incident

Monday morning. 8:45 AM. The phone vibrates.

CRITICAL INFRASTRUCTURE



The Call

No Daily stand-up.
Instead, **200+ people** on
an emergency call. The
sirens are real.

CRITICAL INFRASTRUCTURE



The Scale

Banks are the state's backbone. Failure is not an option; it's a **National Crisis**.



The Call

No Daily stand-up.
Instead, **200+ people** on an emergency call. The sirens are real.



The Regulator

Self-reporting to BaFin.

Every minute of downtime is a formal breach of trust.

THE LOGIN CRASH

Production is at a standstill. DB CPU is locked at **90%**. Response times are drifting from 100ms to **20 minutes**.

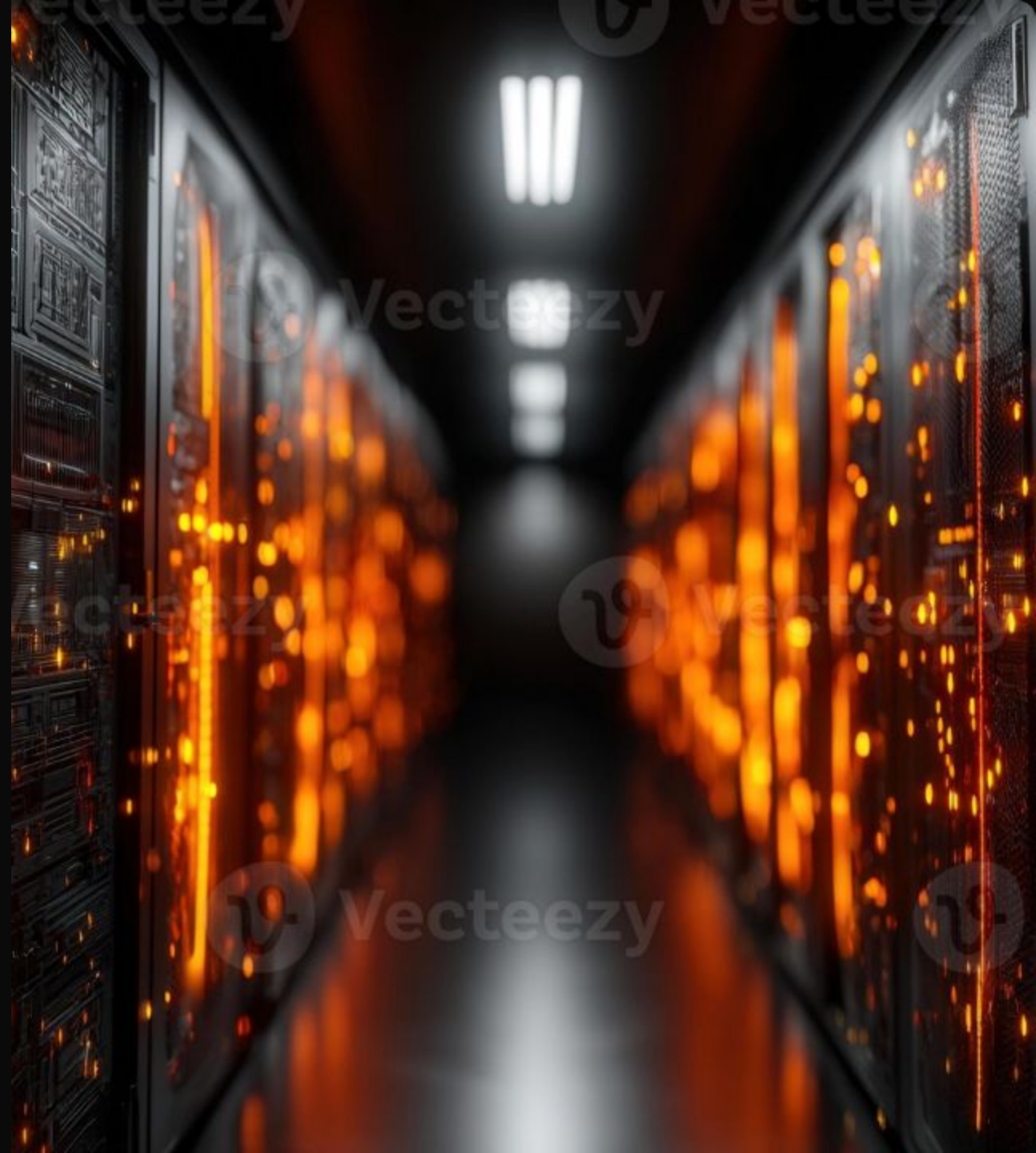


THE LOGIN CRASH

Production is at a standstill. DB CPU is locked at **90%**. Response times are drifting from 100ms to **20 minutes**.

The culprit? A hardcoded timeout in the app layer.

Lesson: **Your DB can be fast, but your application can still fail.**



| THE INFRASTRUCTURE WALL

10B

Total Records

11TB

Oracle Footprint

Exadata

Target Platform

Part II: The Mystery

Why did identical queries perform differently across instances?

THE COLD CACHE TRAP



The Discovery

A Disaster Recovery test was running in the background. We compared environments:

Identical logic, different reality.

THE COLD CACHE TRAP



The Discovery

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Identical logic, different reality.



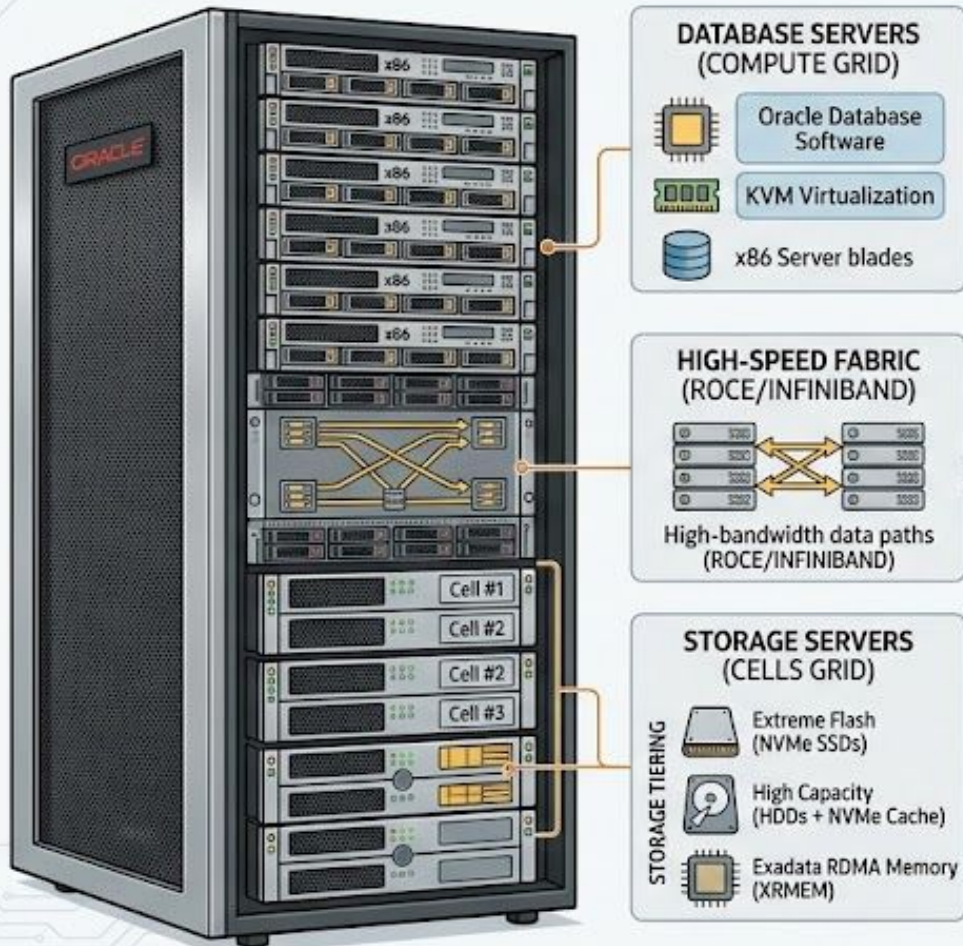
The Lesson

Know your tools. Oracle Exadata Flash Cache wasn't primed. The "Cold Start" was our undoing.

THE COLD CACHE TRAP

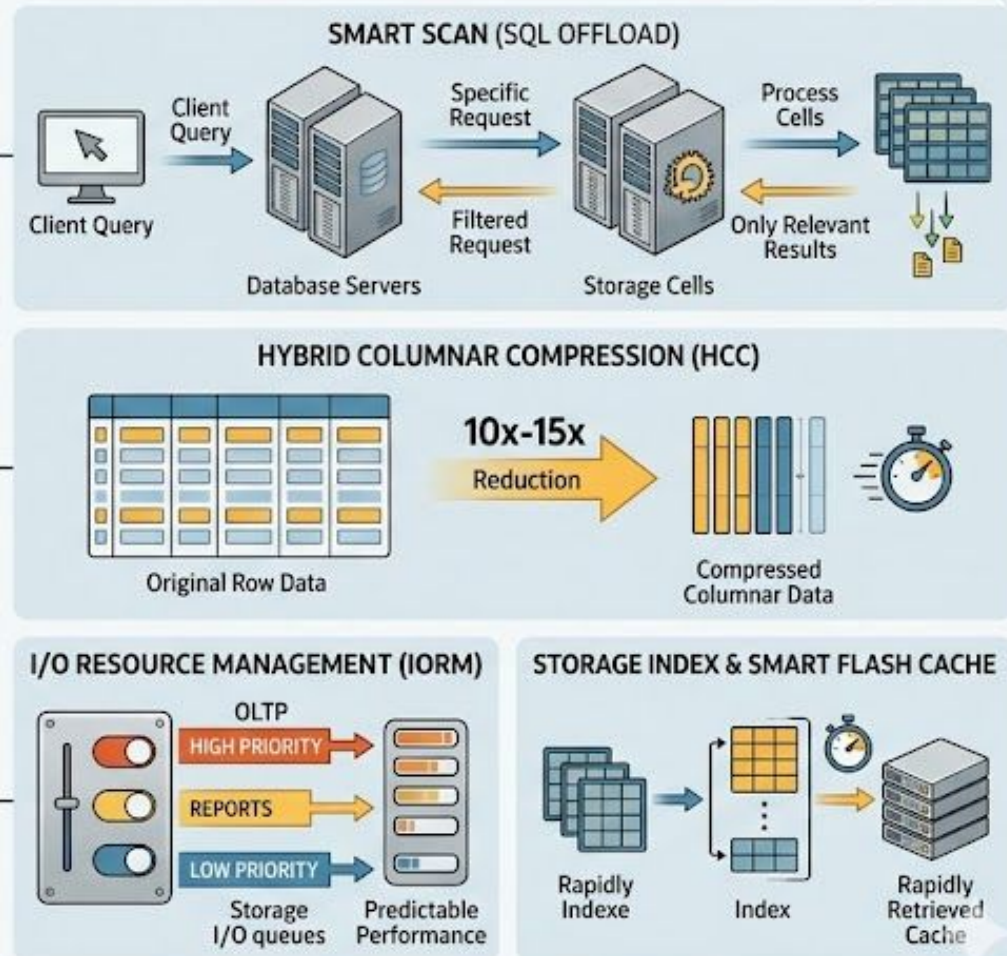
ORACLE EXADATA: ARCHITECTURE & CAPABILITIES

Exadata Structure



Oracle Exadata Database Machine

Exadata Capabilities (Software)



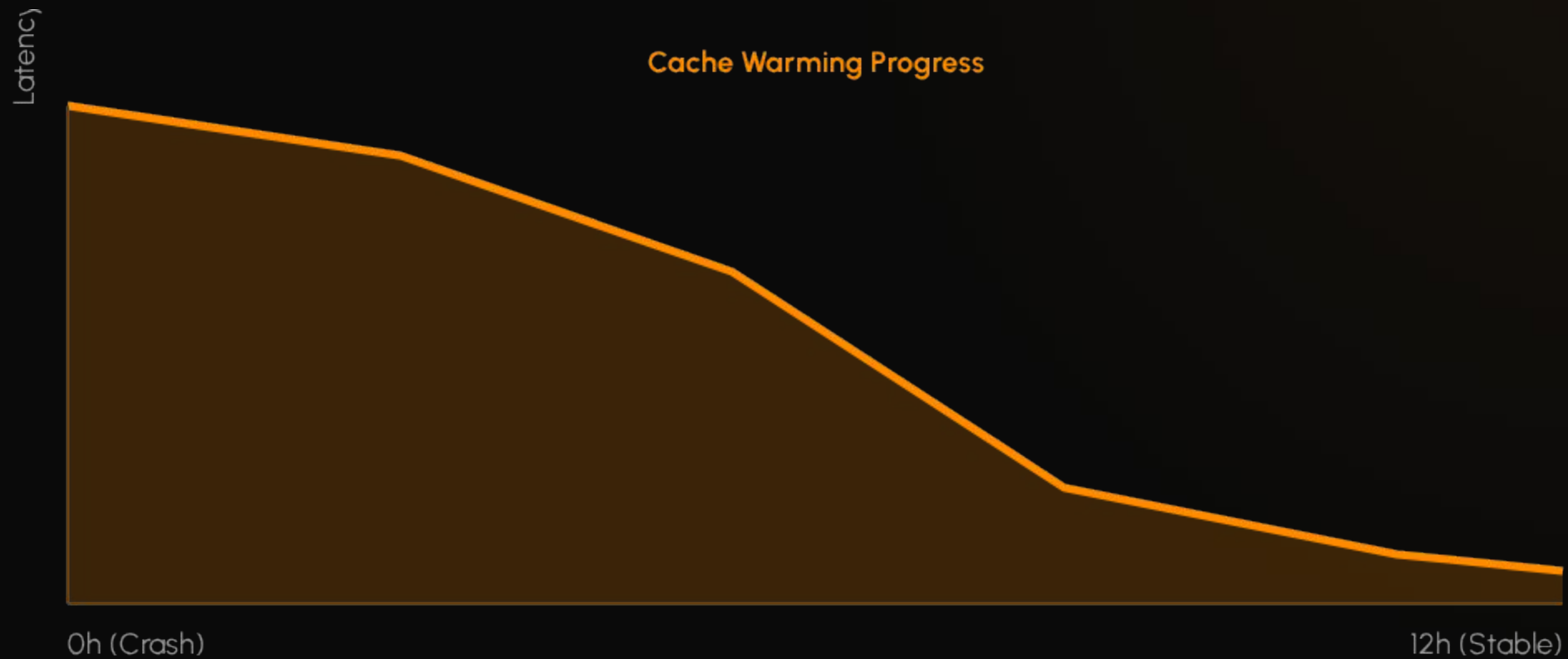
THE STRATEGIC DECISION

| THE STRATEGIC DECISION

*"Do nothing and wait. The cache will
populate itself. Yes, do nothing."*

— *The bravest person in the room*

THE RECOVERY CURVE



As the **Oracle Buffer Cache** warmed, outliers vanished. Patience was the primary fix.

Part III: The Beast SQL

The "Big SQL" that deduplicated 3 tables across 10 billion records.

10 kB of text

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THE Beast SQL – first deduplication

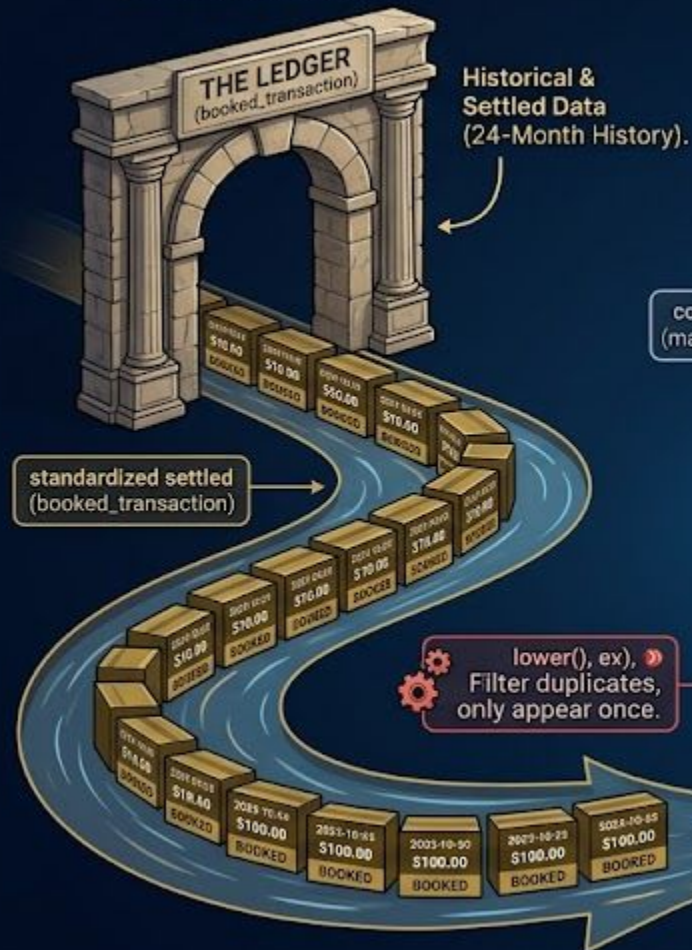
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SELECT
    mbt_filtered.*
FROM
    (
        SELECT
            mbt.*,
            ROW_NUMBER() OVER (
                PARTITION BY
                    mbt_account_nr,
                    mbt_account_currency_code,
                    mbt_transaction_ref_id
                ORDER BY
                    mbt_calculation_ts DESC NULLS LAST
            ) AS rn
        FROM
            marked_booked_transaction mbt
        WHERE
            mbt_account_nr = :1 AND mbt_account_currency_code = :2 AND mbt_valid_until > SYSDATE - 1
    ) mbt_filtered
WHERE
    mbt_filtered.rn = 1
AND mbt_filtered.mbt_transaction_status = 'PENDING'
```

THE Beast SQL – second deduplication

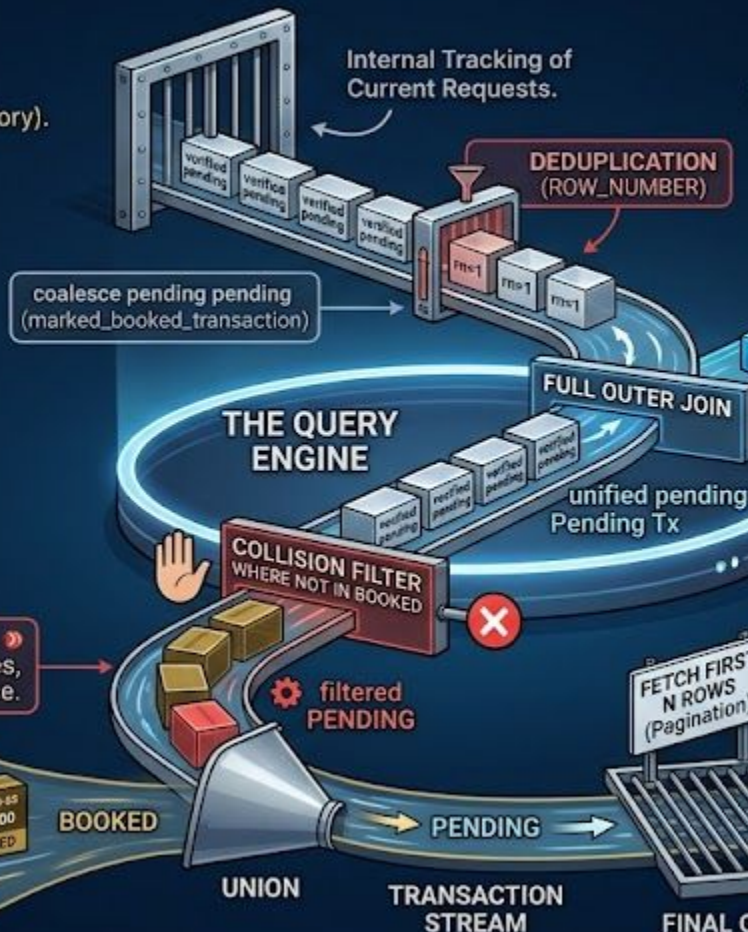
```
) pending_tx
LEFT JOIN (
    SELECT
        tx_ACCOUNT_NR, tx_ACCOUNT_CURRENCY_CODE, tx_BOOKING_DATE,
        tx_BOOKING_TEXT, tx_BOOKING_TEXT_EN, tx_GVO, tx_AMOUNT,
        tx_VALUTA_DATE
    FROM
        booked_transaction
    WHERE
        tx_account_nr = :5 AND tx_account_currency_code = :6 AND
        tx_BOOKING_DATE >= sysdate-10
    AND tx_ACCOUNT_CURRENCY_CODE = :7 ) tx ON pending_tx.
pending_tx_sepa_customer_ref = tx.tx_sepa_customer_ref
```


THE Beast SQL

'PORTAL 1: System of Record' (Booked Transactions)



'PORTAL 2: Internal Markers' (Marked Booked)



'PORTAL 3: Real-Time Feed' (Realtime Transactions)



THE 10-DAY BAND-AID

We found a check removed **3 years ago** during refactoring. We re-added it with a twist.

- Search limited to the **last 10 days**.
- Avoids the 11TB full scan.
- Risk: 7-day Christmas holidays.

It bought us time. The 4-month DR deadline was looming.



THE BIG FIX

Join Tables

Separate **Pending** and **Booked** transactions.
Optimized storage for the active hot-set.

THE BIG FIX

Join Tables

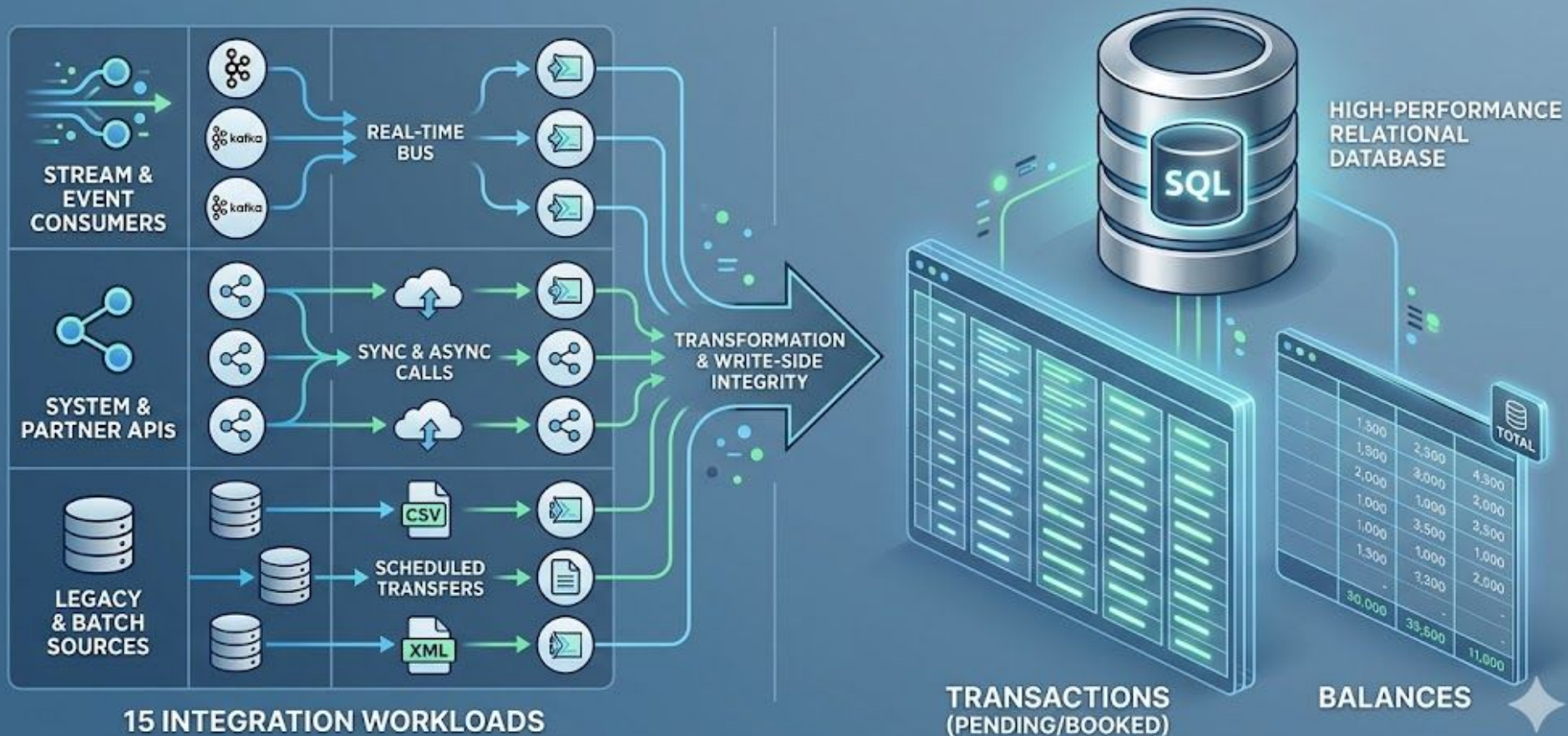
Separate **Pending** and **Booked** transactions.
Optimized storage for the active hot-set.

Write-Side Integrity

Deduplication moved from **Read** to **Write**.
Importers became complex;
Reads became $O(1)$.

THE BIG FIX

TRANSACTION DATA INGESTION: 15 SMARTER IMPORTERS



ARCHITECTURE ROADMAP

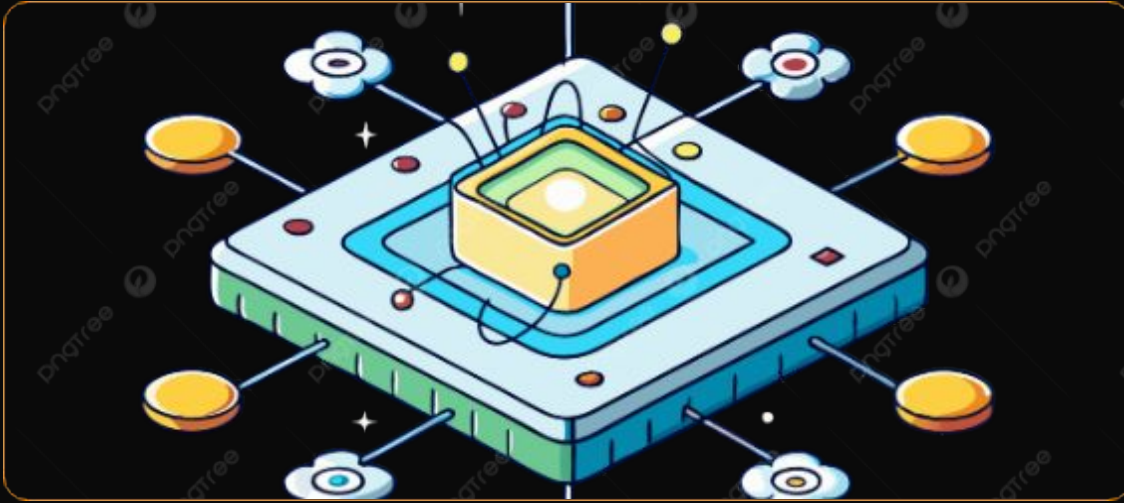


Part IV: The Great Exit



Building the bridge to Cloud PostgreSQL.

THE ZERO-LOSS BRIDGE



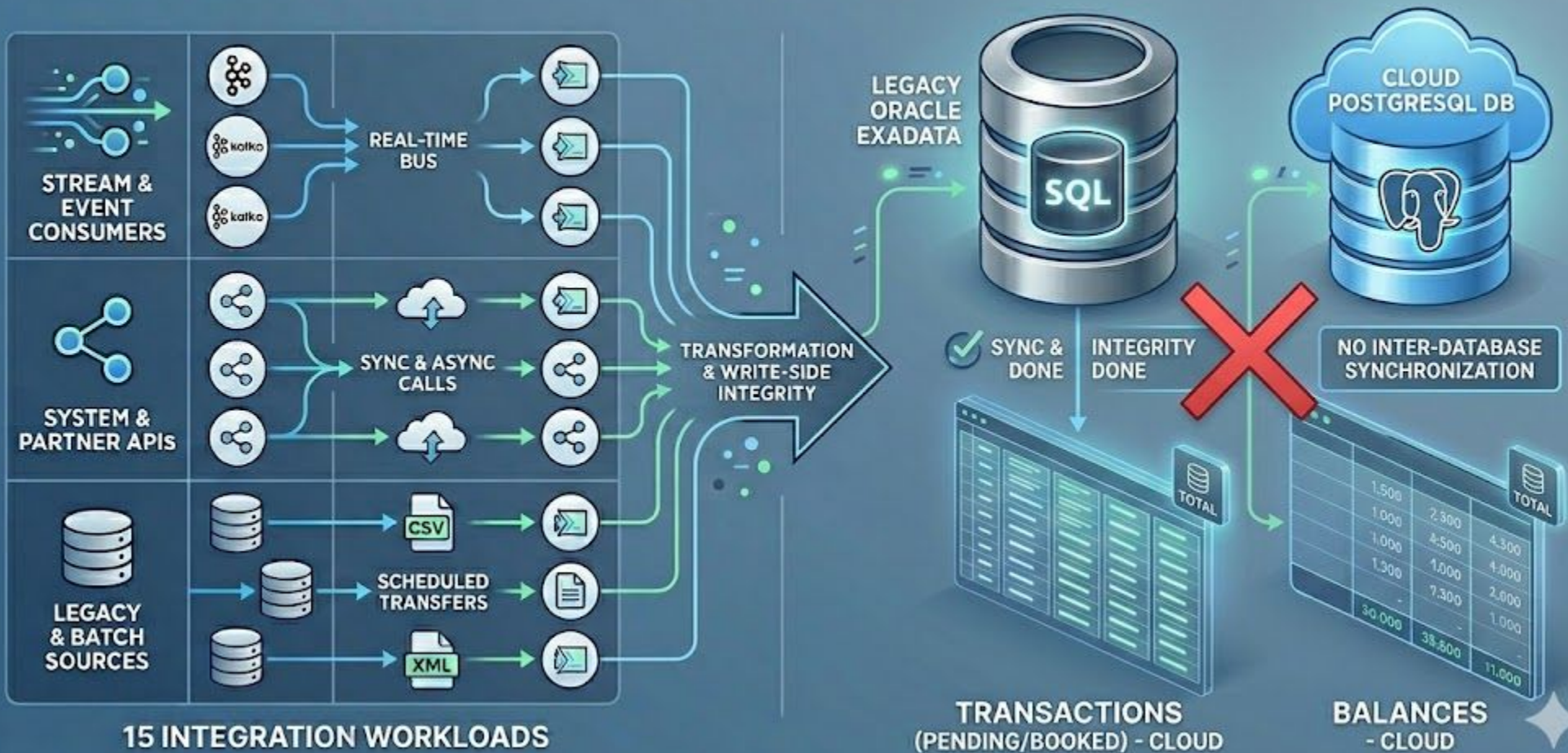
Dual Writes: Our importers store data in Oracle and Postgres simultaneously. No CDC needed.



Migration: Pilot users first. Temporary 2x data overhead. Safe, controlled cutover.

THE ZERO-LOSS BRIDGE

SIMULTANEOUS WRITE-SIDE INGESTION & CLOUD MIGRATION



QUESTIONS?



The Bank Crash Chronicles: Why Our 11TB Oracle Exadata
Architecture Failed

ladecky.com

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